

Climate Change & Its Global Impact
Understanding Our Planet's Greatest Challenge
Environmental Strategy Team - November 2024

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Agenda

Introduction to Climate Change

Root Causes and Greenhouse Effect

Ecosystem Impacts and Biodiversity Loss

Rising Sea Levels and Coastal Threats

Economic Consequences and Vulnerable Sectors

Renewable Energy Solutions

Global Initiatives and Agreements

Case Studies: Regional Impact Examples

Future Predictions and Scenarios

Key Takeaways and Call to Action

Introduction

Climate change refers to long-term shifts in global temperatures and weather patterns

Primary driver is human activity, especially fossil fuel combustion and deforestation

Global average temperature has risen by 1.1°C since pre-industrial times

Current trajectory points to 2.7°C warming by 2100 without intervention

Impacts are already visible across all continents and oceans

Root Causes

Fossil fuel dependence: 73% of greenhouse gases from coal, oil, and gas

Deforestation: 10% of global emissions from land use changes

Industrial agriculture: Methane from livestock, nitrous oxide from fertilizers

Waste management: Landfills produce significant methane emissions

Population growth increasing resource consumption exponentially

Ecosystem Impacts

Polar ecosystems collapsing: Ice loss affects polar bears, penguins

Coral reefs dying: Ocean acidification destroys marine habitats

Amazon rainforest approaching tipping point

Arctic wildlife migration patterns disrupted

Invasive species spreading to new territories

Biodiversity loss accelerating: 1 million species at risk

Forest dieback from droughts and pests

Wetland destruction from sea level rise

Mountain ecosystems shrinking rapidly

Ocean currents disrupted affecting marine life

Rising Sea Levels

Global sea levels rising 3.3mm annually, accelerating to 4.5mm

Thermal expansion accounts for 40% of rise

Ice sheet loss from Greenland and Antarctica contributing 60%

Coastal flooding affects 680 million people currently

By 2100, 590 million people could be displaced from coastal areas

Economic Impact

Annual climate damage costs: \$23 trillion by 2050

Agriculture losses: \$110 billion yearly from droughts and floods

Insurance claims increased 300% in last decade

Healthcare costs rising from heat-related illnesses

GDP impact: 10-18% reduction possible by 2100 in vulnerable nations

Renewable Energy Solutions

Solar capacity grew 20x since 2010, now cheapest electricity source

Wind energy costs dropped 70% in a decade

Battery storage costs fell 89% enabling grid stability

Electric vehicle adoption reaching 14% of new car sales

Green hydrogen emerging for heavy industry and transport

Global Initiatives Progress

Future Projections

1.5°C scenario: 100 million climate migrants by 2050

2°C scenario: 250 million displaced, 3.7 billion in extreme poverty

10°C worst case: Civilization-threatening disruption

Technology pathway: 70% emissions reduction possible with existing tech

Critical decade: Decisions made 2020-2030 determine trajectory

Key Takeaways

Climate crisis is solvable with existing technologies

Immediate action prevents catastrophic warming

Renewables offer economic opportunities, not just costs

Global cooperation essential for meaningful impact

Individual and collective action creates measurable change

Together for a Sustainable Future
Questions & Discussion